

ANALYSIS OF **BMS MEASUREMENT** FUNCTION OF ELECTRIC VEHICLES



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Analysis of BMS Measurement Function of Electric Vehicles

With the fast advancement of electric vehicle technology, the market is expanding rapidly. Consequently, the battery management system (BMS) is becoming increasingly crucial. It ensures the performance and safety of electric cars.

The battery pack is the heart of an electric vehicle. Therefore, its health directly impacts the vehicle's performance and longevity. In this context, the measuring function of the BMS has become critical. It is essential for monitoring and maintaining the battery state.

The Importance of Voltage Monitoring

Voltage monitoring is one of the core tasks of BMS. It impacts the battery's long-term health and performance. It also affects its current state. Mature digital BMS monitors the voltage of individual batteries using high-precision technology. This helps determine the battery's overall health. In addition to traditional direct measuring methods, modern BMS optimizes the voltage monitoring process with algorithms and models. This boosts data quality and dependability.

The Importance of Reading Rate and Application Differences

The read rate is crucial for BMS. It directly affects real-time data and the system's response speed. A low reading rate is sufficient in applications with low dynamic changes, such as a backup power supply. **For example, 1 minute or 10 seconds per point is adequate. However, a high reading rate is crucial in scenarios where the current changes frequently, such as in electric vehicles. A rate of 1 point per second is necessary.**

In the field of research, researchers may require a reading rate of dozens of points per second. This high-speed reading provides more accurate data when battery performance changes rapidly. It helps optimize the performance and response of the entire vehicle.

The Impact of Data Accuracy

Data accuracy is another key indicator for measuring BMS performance. In basic application scenarios like full battery charging or full discharge detection, an accuracy of 100mV is sufficient.

However, in more advanced applications, higher accuracy is particularly important. Scenarios requiring even battery pack charging need an accuracy of 10-30mV. Accurate data not only helps improve the efficiency and lifespan of batteries but also reduces safety risks to a certain extent.